



SRI RAMACHANDRA

INSTITUTE OF HIGHER EDUCATION AND RESEARCH

(Category - I Deemed to be University) Porur, Chennai

SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

**A GAMIFIED WEB-BASED ASSESSMENT PLATFORM FOR
INTERACTIVE NURSING EDUCATION**

CYB23IN201 – INTERNSHIP II

PROJECT REPORT

Submitted by

G L SWAMINATHAN – E0224033

In partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

(Cyber Security & Internet of Things)

Sri Ramachandra Faculty of Engineering and Technology

Sri Ramachandra Institute of Higher Education and Research, Porur, Chennai -600116

JULY, 2026



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BONAFIDE CERTIFICATE

Certified that this project report “**A GAMIFIED WEB-BASED ASSESSMENT PLATFORM FOR INTERACTIVE NURSING EDUCATION**” is the bonafide record of work done by “**G L SWAMINATHAN – E0224033**” who carried out the internship work under my supervision.

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Evaluation Date:

INTERNAL EXAMINER

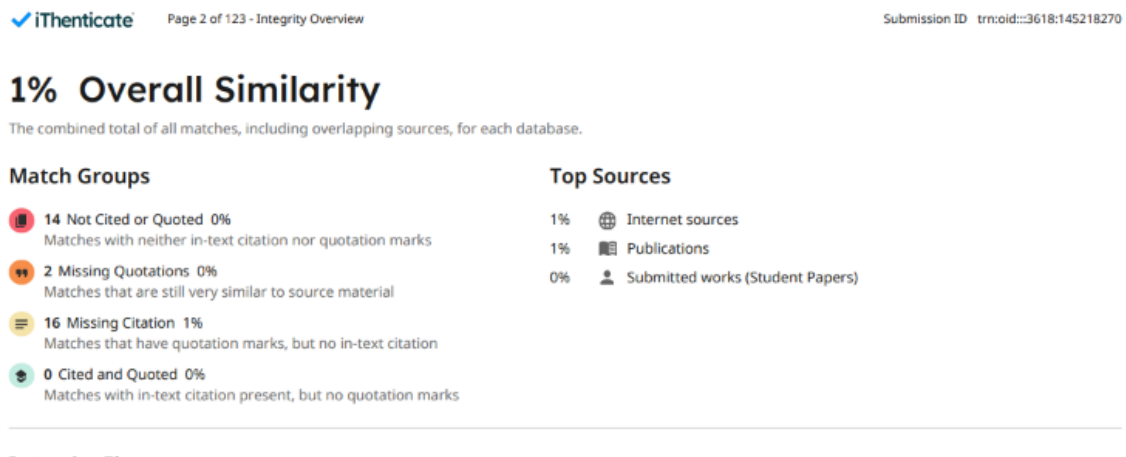
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Signature of the Internship Coordinator

Signature of the Supervisor



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ABSTRACT

The rapid evolution of modern healthcare systems and the increasing complexity of clinical environments demand a high level of preparedness, clinical reasoning, and quick decision-making from nursing professionals. From an educational standpoint, this paradigm shifts student cognitive interaction from simple recognition to deep conceptual retrieval. This active engagement reinforces learning outcomes and allows students to retain complex medical information over a longer duration. Traditional pedagogical methods in nursing education, which heavily rely on didactic lecturing, passive reading, and static multiple-choice question sheets, often fail to stimulate active recall and fail to prepare students for high-stakes, real-world clinical environments. Pedagogical studies confirm that immediate feedback loops reduce cognitive load and improve exam outcomes in medical curricula. By showing detailed explanations after each question, students can learn from mistakes dynamically. This study addresses these educational limitations by designing, developing, and evaluating NurseQuest, a gamified web-based education and assessment platform specifically tailored to the nursing curriculum. This research highlights the importance of visual and auditory evaluations, which are crucial for assessing clinical readiness. The inclusion of auscultation audio and ECG diagrams ensures students are tested on multi-sensory diagnostics. Built using a modern full-stack web architecture—comprising a React 19 single-page application frontend, a Node.js Express 5 backend API server, and a stateful Socket.IO real-time communication layer powered by a Supabase PostgreSQL database—NurseQuest bridges the gap between traditional learning and interactive engagement. The full-duplex communication channel maintains low-latency data streams, ensuring real-time synchronized student response loops. This creates a fair, competitive, and high-energy learning environment in the classroom. The platform implements a unified gamification engine featuring experience points (XP), a 7-tier nursing rank progression system, active streaks, and milestone achievements to motivate continuous learning. The relational storage structure enforces strict validation rules, preventing schema anomalies and data inconsistency. By maintaining clean referential integrity, the database supports stable transactional audits of student activities. Crucially, the platform supports nine specialized, clinically relevant question formats, including audio-based auscultation challenges, image-based ECG waveform recognition, sequence ordering for medical procedures, and slider-based vital sign estimations, moving far beyond standard text-only multiple-choice trivia.